

ADIA : Lab

Causal Factor Analysis is a Necessary Condition for Investment Efficiency

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Seminar's Objective

- Choosing a correct model specification requires knowledge of the causal graph that characterizes the underlying data-generating process (DGP)
- The econometric canon favors models that over-control for colliders, because these models
 - always exhibit higher R-squared; and
 - exhibit, under general conditions, higher adjusted R-squared and lower p -values
- A misspecified statistical model cannot attribute risks and returns correctly, hence leading to inefficient investment allocations
- **Conclusion: It is not possible to estimate the efficient frontier without a causal factor model**

The full paper is available here: https://ssrn.com/abstract_id=5131050

The Investment Allocation Problem

Factor Model

- Consider N assets, with returns $X_1, \dots, X_N$, and K risk factors, with returns $F_1, \dots, F_K$, related by the equation.

$$X = BF + \varepsilon$$

where $B \in \mathbb{R}^{N \times K}$ are the exposures, and $\varepsilon \in \mathbb{R}^N$ are the asset-specific returns. The variance-covariance matrix of X can be written as

$$V = \text{Var}[X] = B\Sigma B' + \Delta$$

$$\Sigma = \text{Var}[F]$$

$$\Delta = \text{Var}[\varepsilon]$$

Investment Strategy (1/2)

- Consider an investor who wishes to compute the optimal portfolio ω with minimal variance subject to achieving certain factor exposures. Formally,

$$\begin{aligned} \min_{\omega} \quad & \omega' V \omega \\ \text{s. t. : } & \omega' B = c \end{aligned}$$

where ω has size $N \times 1$, c has size $1 \times K$, and V is the $N \times N$ positive-definite variance-covariance matrix.

- The closed form solution to the above optimization problem is

$$\omega^* = V^{-1} B A^{-1} c'$$

$$A = B' V^{-1} B$$

Investment Strategy (2/2)

- In particular, when $N = K$, the solution reduces to

$$\omega^* = B'^{-1}c'$$

- For illustrative purposes, and without loss of generality, consider the case of two assets and two factors, where the investor wishes to have unit exposure to F_2 , while hedging its exposure to F_1 , thus $K = 2$, $N = 2$ and $c = [0 \quad 1]$
- For our analysis, it is also helpful to derive the solution for the case where $K = 1$, $N = 2$ and $c = 1$, which can be expressed as

$$\omega^* = \frac{V^{-1}B}{B'V^{-1}B}$$

Factor Model Misspecification

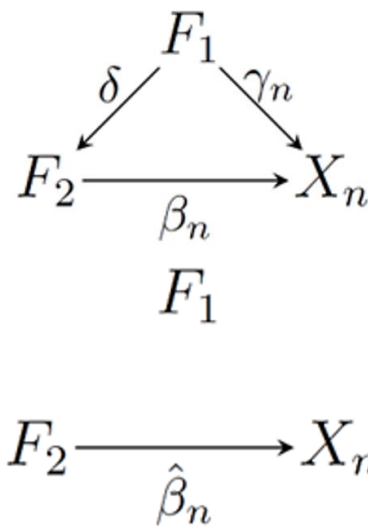
Case 1: Failing to Control for a Confounder

- When the wrong backdoor adjustment set is chosen, the factor model will be misspecified, resulting in biased estimators
- First, consider the case where a researcher fails to control for a confounder, as illustrated in the DAG on the right. The true (left) and misspecified (right) matrices of exposures are

$$B_{true} = \begin{bmatrix} \gamma_1 & \beta_1 \\ \gamma_2 & \beta_2 \end{bmatrix} \quad B_{miss} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix}$$

- [López de Prado and Zoonekynd \[2024\]](#) show that, (assuming unit variances) for asset $n \in \{1,2\}$, β_n will be estimated as

$$\hat{\beta}_n = \beta_n + \frac{\gamma_n \delta}{1 + \delta^2}$$



A researcher misjudges the top causal graph (ground truth) to be the causal graph at the bottom.

Case 2: Controlling for a Collider

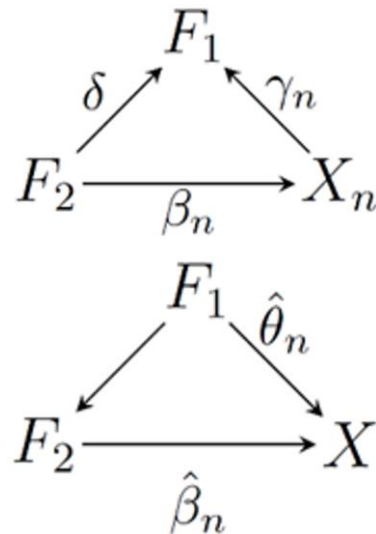
- Second, consider the case where a researcher controls for a collider, as illustrated on the right. The true (left) and misspecified (right) matrices of exposures are

$$B_{true} = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} \quad B_{miss} = \begin{bmatrix} \hat{\theta}_1 & \hat{\beta}_1 \\ \hat{\theta}_2 & \hat{\beta}_2 \end{bmatrix}$$

- López de Prado and Zoonekynd [2024] show that, (assuming unit variances) for asset $n \in \{1,2\}$, β_n and θ_n will be estimated as

$$\hat{\beta}_n = \beta_n - \gamma_n \delta$$

$$\hat{\theta}_n = \frac{\gamma_n}{1 + \gamma_n^2}$$



A researcher misjudges the top causal graph (ground truth) to be the causal graph at the bottom.

Investment Inefficiency

Case 1: Failing to Control for a Confounder

- In the first case, the optimal allocations are

$$\omega_{true}^* = B_{true}'^{-1}c' = \frac{1}{\gamma_1\beta_2 - \gamma_2\beta_1} \begin{bmatrix} -\gamma_2 \\ \gamma_1 \end{bmatrix}$$

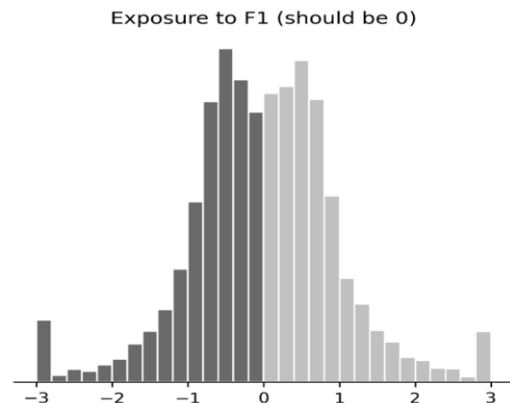
$$\omega_{miss}^* = \frac{V^{-1}B_{miss}}{B_{miss}'V^{-1}B_{miss}}$$

$$B_{miss}' = \begin{bmatrix} \beta_1 + \frac{\gamma_1\delta}{1+\delta^2} & \beta_2 + \frac{\gamma_2\delta}{1+\delta^2} \end{bmatrix}$$

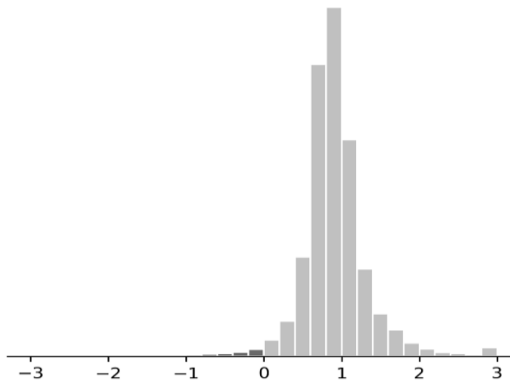
- The solution from the misspecified model is inefficient, because it exposes the investor to unwanted risks:

$$\omega_{miss}^* B_{true} = \omega_{miss}^* \begin{bmatrix} \gamma_1 & \beta_1 \\ \gamma_2 & \beta_2 \end{bmatrix} \neq \begin{bmatrix} 0 & 1 \end{bmatrix}$$

- On the right, exposures for a wide range of parameters



Exposure to F2 (should be 1)



Case 2: Controlling for a Collider

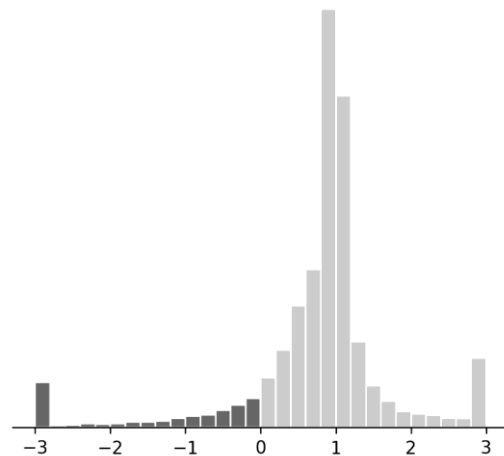
- In the second case, the optimal allocations are

$$\begin{aligned}
 B_{true}' &= [\beta_1 \quad \beta_2] \\
 \omega_{true}^* &= \frac{V^{-1} B_{true}}{B_{true}' V^{-1} B_{true}} = \frac{1}{\beta_1^2 + \beta_2^2} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} \\
 \omega_{miss}^* &= \frac{B_{miss}'^{-1} c'}{1} \\
 &= \frac{1}{\beta_2 \gamma_1 (1 + \gamma_2^2) - \beta_1 \gamma_2 (1 + \gamma_1^2) + \gamma_1 \gamma_2 (\gamma_1^2 - \gamma_2^2) \delta} \begin{bmatrix} -\gamma_2 (1 + \gamma_1^2) \\ \gamma_1 (1 + \gamma_2^2) \end{bmatrix}
 \end{aligned}$$

- The solution from the misspecified model is inefficient, because it exposes the investor to unwanted risks:

$$\omega_{miss}^* B_{true} = \omega_{miss}^* \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} \neq 1$$

Exposure to F2 (should be 1)



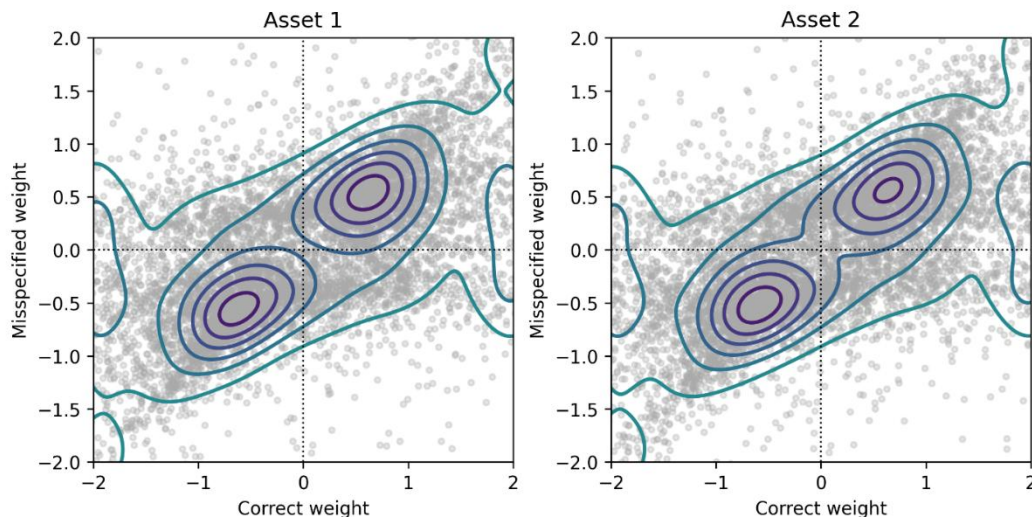
Exposures for a wide range of parameters, for a wide range of 10,000 parameter combinations, drawn from a uniform distribution bounded in $[-1.5, 1.5]$.

Investment Losses Caused By Misspecification

Case 1: Failing to Control for a Confounder

- Furthermore, applying a wrong backdoor adjustment set leads to “bad” strategies, where investors buy what they should have sold, and vice versa, even when investors have perfect knowledge of the mean and covariances.
- The figure below plots pairs $(\omega_{true}^*, \omega_{miss}^*)$ when the model fails to control for a confounder

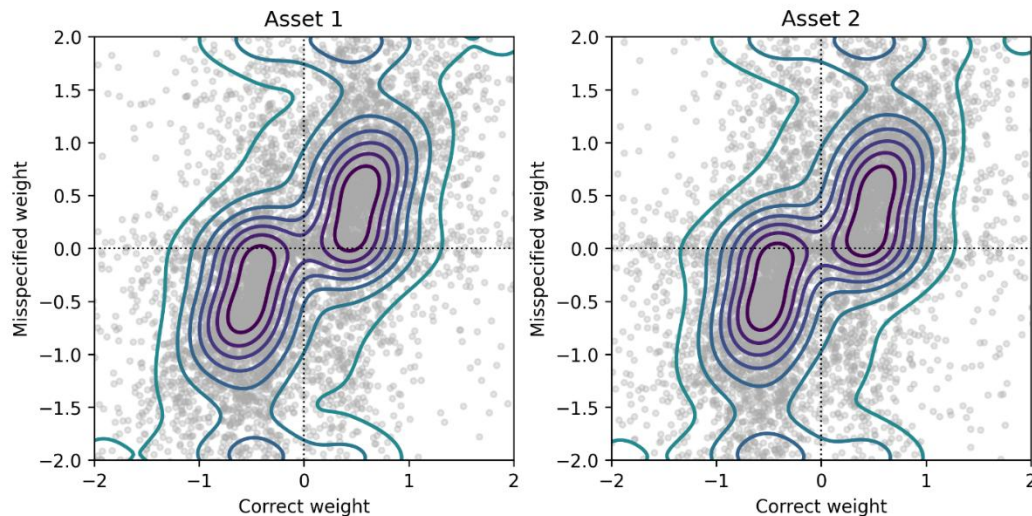
- For instance, when $\beta_1 = \beta_2 = 1$, $\delta = -0.5$, $\gamma_1 = 1$, $\gamma_2 = 1.5$, then $\omega_{true}^* = (3, -2)$, whereas $\omega_{miss}^* = (1.2, 0.8)$



Case 2: Controlling for a Collider

- The second case is particularly dangerous, because it includes scenarios where there is a reversal of *all* signs
- The figure below plots pairs $(\omega_{true}^*, \omega_{miss}^*)$ when the model control for a collider

- For instance, when $\beta_1 = \beta_2 = 1$, $\delta = 1.5$, $\gamma_1 = 1.5$, $\gamma_2 = -0.5$, then $\omega_{true}^* = (0.5, 0.5)$, whereas $\omega_{miss}^* = (-5, -3)$



Generalization

- Our analysis has focused on the computation of the optimal portfolio in the case where $K = 2$, $N = 2$ and $c = [0 \ 1]$
- Increasing the number of assets, factors or constraints will make the calculations more cumbersome, only to reach the same conclusions
- In fact, the larger the number of assets, factors and constraints, the greater the potential for inefficiency and losses is, because making the correct risk attribution becomes more challenging

The Econometric Canon

Explanatory Power & p -Values

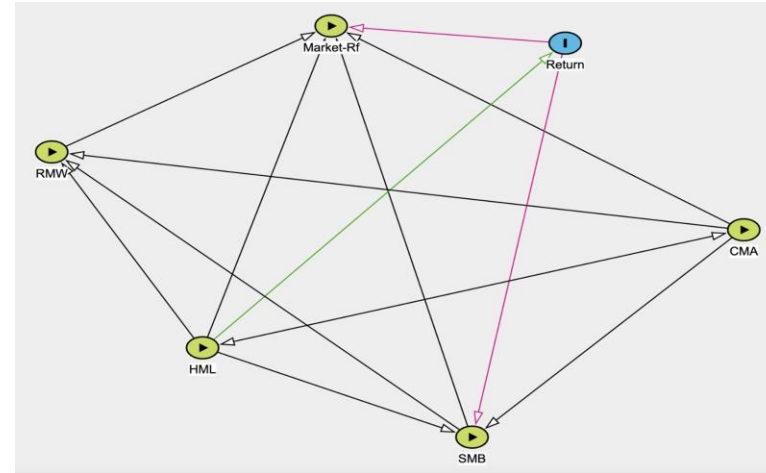
- The R-squared of a factor model that over-controls for a collider always exceeds the R-squared of the correct model
 - Given enough observations, the adjusted R-squared of a factor model that over-controls for a collider exceeds the adjusted R-squared of the correct model
- When the collider effects are strong and of similar magnitude, or the causal effect is weak, the p -values of estimated coefficients in a factor model that over-controls for a collider are smaller than those of the correct model
 - The presence of colliders facilitates p -hacking under rather general conditions

$$\begin{aligned} R_{\text{correct}}^2 &= 1 - \frac{\text{Var}[u]}{\text{Var}[Y]} \\ &= 1 - \frac{1}{1 + \beta^2} \\ R_{\text{over-controlled}}^2 &= 1 - \frac{\text{Var}[\varepsilon]}{\text{Var}[Y]} \\ &= 1 - \frac{1}{(1 + \gamma^2)(1 + \beta^2)} \end{aligned}$$

Controlling for collider Z introduces non-causal association, with the effect of inflating the R-squared and, given enough observations, also the adjusted R-squared.

Canonical Model Selection

- The econometric canon is seemingly ignorant of the problem of collider bias
 - Greene [2012, p. 136] indicates that the downside of including irrelevant variables is “the reduced precision of the estimates,” and that “[o]mitting variables from the equation seems generally to be the worse of the two errors”
 - Greene [2012, p. 137] recommends a “general-to-simple” approach, where a researcher starts with a “kitchen sink regression, which contains every variable that might conceivably be relevant”
 - Fama and French [2015] argument for adding two factors to their initial three-factor model specification was that “the five-factor model performs better than the three-factor model when used to explain average returns”

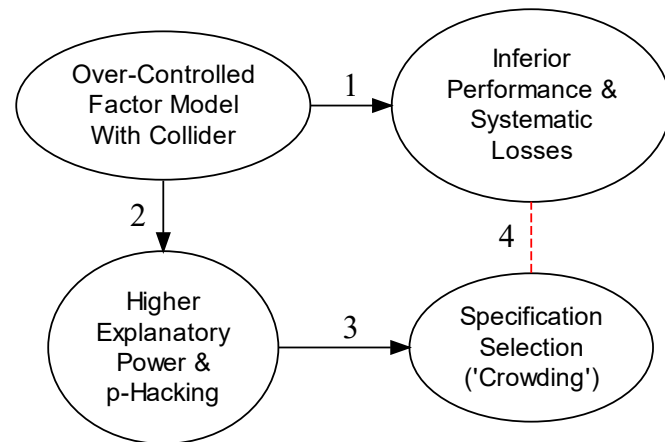


[Sadeghi et al. \[2024\]](#) apply the PC algorithm for causal discovery to the variables used in Fama and French [1993], and find that the HML factor may be a collider.

[Gu et al. \[2023\]](#) find that Fama and French [2015] may also be misspecified, with the presence of at least one collider (market returns net of the risk-free rate).

The Causal Mechanism for Adverse Outcomes

1. Strategies based on misspecified factor models can yield systematic losses in the collider case
2. Misspecified factor models that over-control for a collider exhibit a higher R-squared than correctly specified models
 - Under certain conditions, colliders make it easier to *p*-hack a factor
3. Canonical approaches for specification-selection in econometrics favor over-controlled models, crowding out correctly specified models



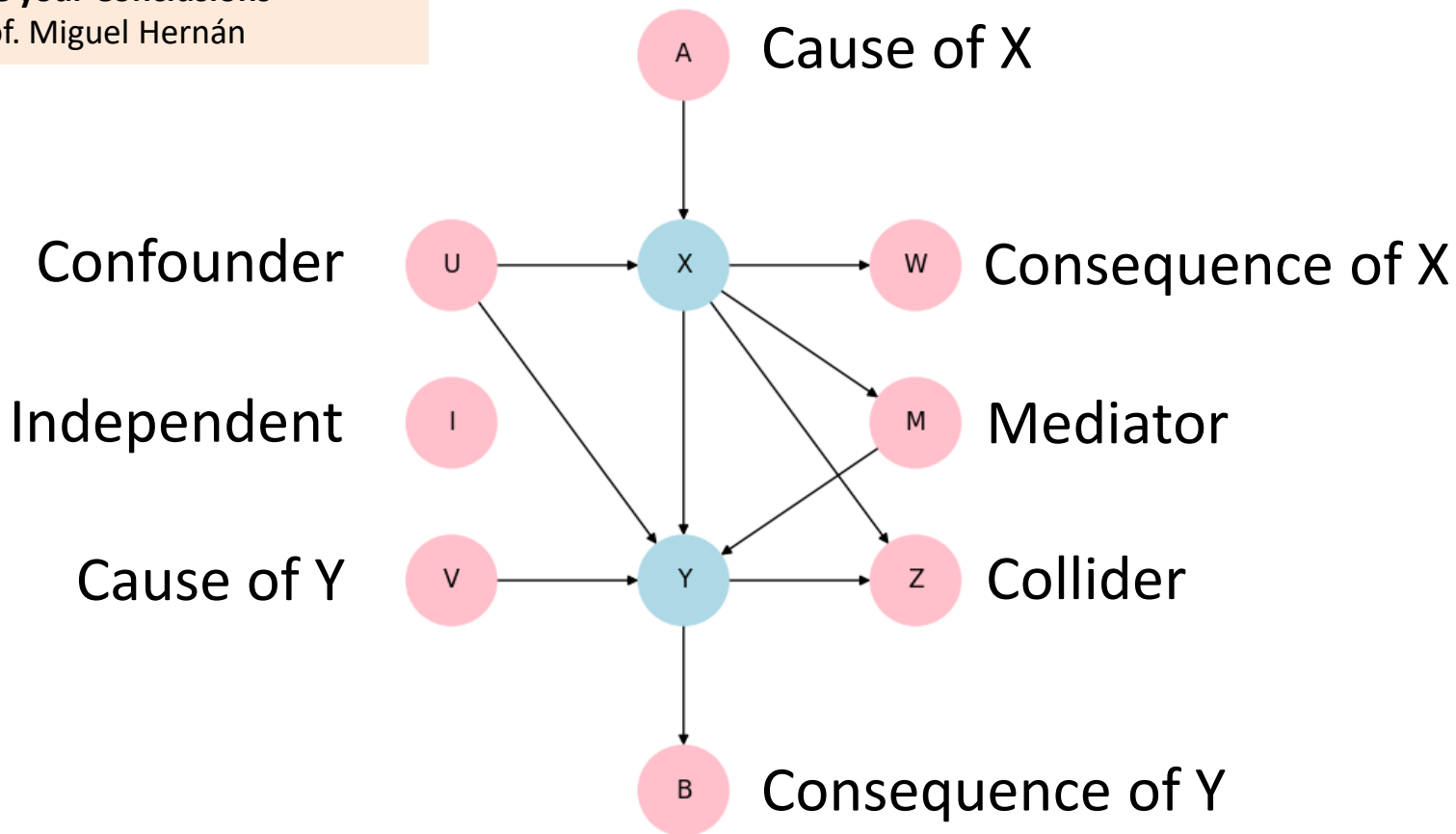
The combined effect of (1,2,3) is an increased association between selected (either published or deployed) factor investing strategies and underperformance, including systematic losses.

This causal mechanism provides a channel for “few-shot *p*-hacking.”

Solutions

**“Draw your Assumptions
before your Conclusions”**

Prof. Miguel Hernán



Causal Discovery Algorithms

- Over the past three decades, statisticians have developed numerous computational methods and algorithms for the discovery of causal relations, represented as directed acyclic graphs (DAGs)
- These methods can be divided into three types:
 - **Score-based algorithms** optimize a defined score function, and can be used in the absence latent confounders (e.g., GES)
 - **Constraint-based methods** exploit conditional independence relationships in the data (e.g., PC, FCI)
 - FCI can give asymptotically correct results even if there are latent confounders
 - **Functional causal models** (FCMs) distinguish between DAGs in the same equivalence class



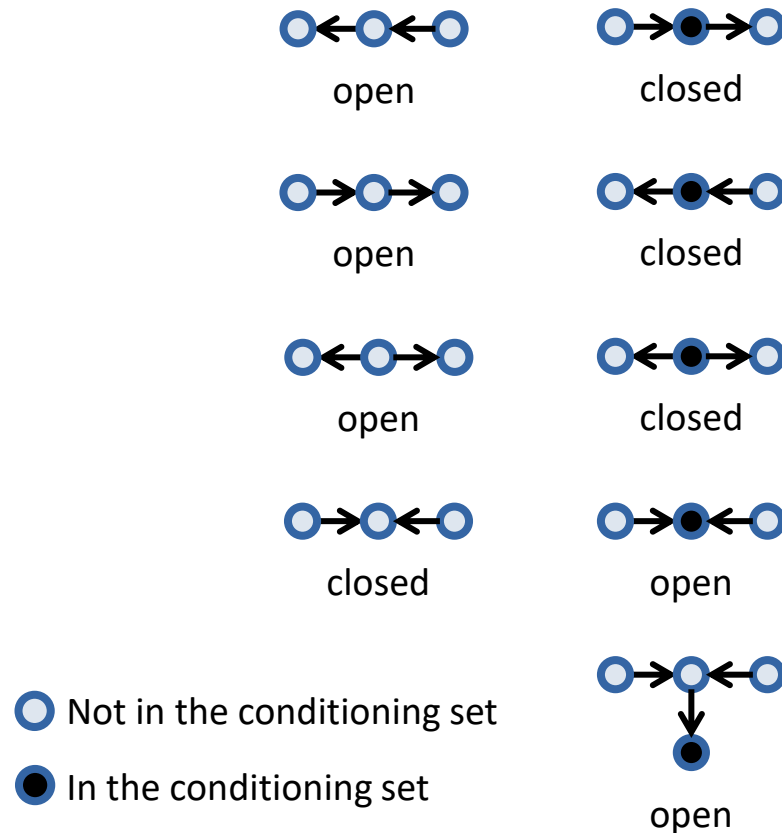
Abstract

Previous asymptotically correct algorithms for recovering causal structure from sample probabilities have been limited even in sparse causal graphs to a few variables. We describe an asymptotically correct algorithm whose complexity for fixed graph connectivity increases polynomially in the number of vertices, and may in practice recover sparse graphs with several hundred variables. From sample data with $n = 20,000$, an implementation of the algorithm on a DECStation 3100 recovers the edges in a linear version of the ALARM network with 37 vertices and 46 edges. Fewer than 8% of the undirected edges are incorrectly identified in the output. Without prior ordering information, the program also determines the direction of edges for the ALARM graph with an error rate of 14%. Processing time is less than 10 seconds. *Keywords* DAGs, Causal Modelling.

Discovering a causal graph typically requires extra-statistical (beyond observational) information, in the form of an experiment or expert knowledge. However, under some assumptions, it is possible to derive causal graphs that are *consistent with* the observed sample.

Specification Selection (1/2)

- **The sufficient conditioning set is not necessarily unique**
 - For instance, the parents of X may form a sufficient conditioning set for $X \rightarrow Y$, but it may be needlessly large
- More generally, a sufficient conditioning set is a set of nodes blocking all the non-causal paths from X to Y , while leaving all the causal paths open
- If some variables are not observed, things get more complicated, but do-calculus can tell us if the effect can be estimated from observational data alone, and how

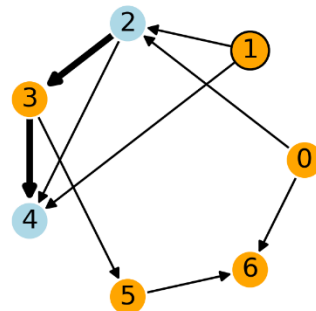


Specification Selection (2/2)

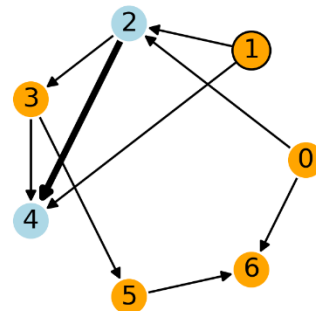
To assess the strength of the causal relation

$X \rightarrow Y$:

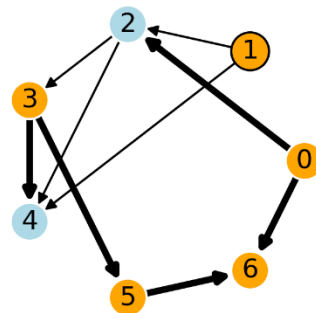
- List all the (undirected) paths from X to Y
- All the non-causal paths should be blocked
 - if not, condition on one or more nodes to block them
- All the causal paths should be open
 - if not, adjust the conditioning set to unblock them



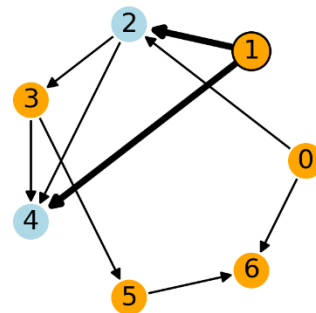
Causal path



Causal path



Blocked by the
collider 6



Blocked if we
condition on 1

The Dawn of Causal Factor Investing

- A scientific theory is a falsifiable statement of the form “ X causes Y through mechanism M ”
- Scientific theories matter to investors because
 - causality is a necessary condition for portfolio efficiency
 - associational models misattribute risks and performance, thus preventing investors from building efficient portfolios
 - causal models enable counterfactual reasoning, hence the stress-testing of investment portfolios in a coherent and forward-looking manner
 - associational models cannot answer counterfactual questions, such as what would be the effect of Y on a not-yet-observed scenario X , thus exposing those relying on associations to black-swan events
- **Financial economists’ adoption of causal inference has the potential to transform investing into a truly scientific discipline**

Type	Rigor	Example
Randomized controlled trials	Very high	Algo-wheel experiments
Natural experiments	High	Market-maker reaction to random spikes in order imbalance
Simulated interventions	Medium	Estimate effect of HML using a causal graph
Econometric (observational) studies	Low	Factor investing literature; backtested investment strategies
Case studies	Very low	Broker report / analysis
Expert opinion	Anecdotal	Investment guru’s prediction

Hierarchy of evidence: Pre-scientific vs. scientific evidence in financial research.

For More Information



Download for free

Causal Factor Investing

(Cambridge University Press, 2023)

Available at:

<https://www.cambridge.org/core/elements/causal-factor-investing/9AFE270D7099B787B8FD4F4CBADE0C6E>

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